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DIQ_Project-Dataset 10

[056490] DATA AND INFORMATION QUALITY

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Abstract

This project presents the complete design, implementation, and validation of a Data Quality Engineering Pipeline applied to the "**Comune di Milano**" open dataset **Pubblici Esercizi Fuori Piano**. Starting from an extensive exploratory assessment, we uncovered a heterogeneous set of issues spanning syntactic irregularities, malformed text encodings, semantic inconsistencies, missing or ambiguous attributes, and domain specific violations.

Our approach integrates multiple classes of techniques: artefact detection and correction for misencoded characters, systematic extraction of structured fields from free-text location descriptions, categorical domain normalization, and transaction-based decomposition of the complex **Settore storico pe** attribute followed by automatic consolidation of redundant labels through a combination of **Jaro-Winkler** similarity and phonetic distance. In addition, we incorporate external geographical knowledge through **GeoCoding** (for **Ubicazione**) services to resolve inconsistent municipality identifiers, enabling a principled reconciliation of spatial attributes.

The resulting dataset is not only structurally and semantically aligned, but also significantly more complete, accurate, and internally consistent. By transforming a noisy open dataset into a robust, analysis-ready resource, the work demonstrates how formal Data & Information Quality methodologies can be operationalized into a coherent and reproducible pipeline capable of supporting downstream analytical and machine-learning tasks.

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1 | SetUp Choices: Methodological Setup and Design Decisions

1.0.1. Computational Environment and Toolchain

The project is implemented in **Python**, which provides a flexible and expressive environment for data quality assessment and data preparation tasks. The core of the implementation relies on **pandas**, selected for its ability to support fine-grained data inspection and manipulation at both column and cell level. This capability is essential when evaluating quality dimensions such as completeness, accuracy, consistency, and uniqueness. **NumPy** is used to support numerical operations, domain checks, and type enforcement, particularly for attributes requiring numerical validation.

The overall environment has been designed to ensure reproducibility and transparency, allowing each transformation step to be explicitly traced and justified.

1.0.2. Data-Driven Design Decisions

An initial exploratory analysis highlighted the heterogeneous nature of the dataset. While some attributes are well-structured and domain-constrained (e.g., identifiers, numerical surfaces, categorical codes), others embed multiple pieces of information within semi-structured or free-text fields, such as **Ubicazione** and **Settore storico pe**.

This observation played a central role in guiding the data preparation strategy. Instead of applying uniform cleaning rules, different techniques were selected depending on the semantic role and structural complexity of each attribute.

1.0.3. Rule-Based Extraction from Textual Attributes

For attributes containing structured information encoded as text, a rule-based extraction strategy was adopted. Regular expressions were employed to recover latent fields such as civic numbers, access types, block identifiers, and municipality codes directly from textual descriptions. This approach allows the recovery of meaningful information while preserving the original semantics, and it avoids information loss that would result from coarse-grained cleaning.

The choice of regex-based extraction is motivated by the deterministic nature of the patterns observed during exploration and by the need for interpretable and reproducible transformations.

1.0.4. Semantic Harmonization of Categorical Attributes

Several categorical attributes exhibited inconsistencies due to abbreviations, spelling variations, and syntactic noise. To address this issue, string similarity techniques were employed to identify semantically equivalent categories. In particular, the **Jaro–Winkler** similarity measure was selected because of its effectiveness in handling typographical variations while remaining sensitive to meaningful differences in short strings.

This similarity-based consolidation was applied conservatively, combining automated detection with frequency-based preservation rules to minimize the risk of semantic over-merging.

1.0.5. Encoding Multi-Valued Categorical Information

The **Settore storico pe** attribute encodes multiple sector categories within a single textual field. Rather than forcing a single-label representation, the attribute was modeled as a multi-valued categorical variable. The raw textual values were tokenized and transformed into transactions, which were then encoded using a **TransactionEncoder** to produce a **Boolean Feature Matrix**.

This design choice preserves the full informational content of the attribute, improves interpretability, and aligns with established best practices for handling composite categorical data in data quality and analytics pipelines.

1.0.6. Geographic Validation through External Sources

In cases where internal constraints were insufficient to resolve inconsistencies particularly for geographic attributes external knowledge was incorporated through the **Nominatim GeoCoding** service based on **OpenStreetMap**. This resource was used selectively to validate and disambiguate municipality codes, rather than as a primary data source.

The use of external data was carefully controlled, acknowledging the dynamic nature of open geographic databases while leveraging them to support informed consistency checks.

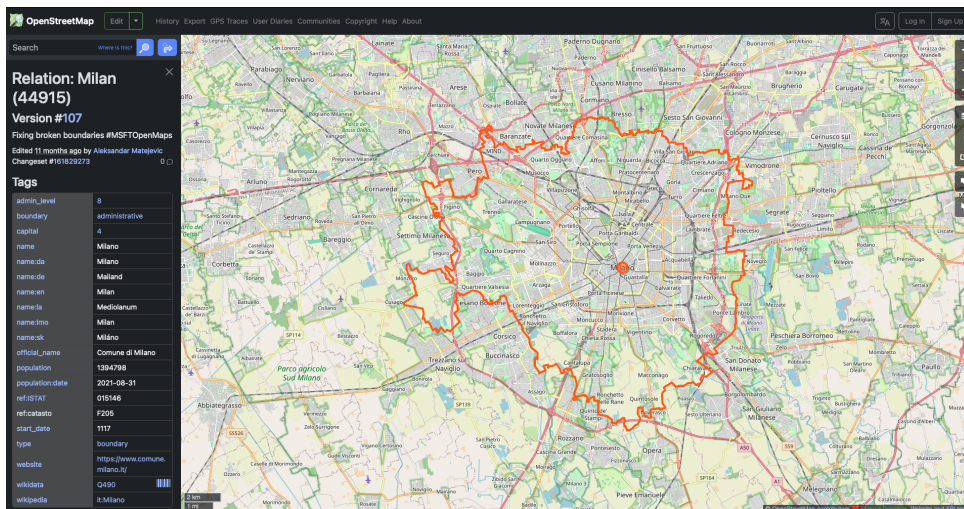


Figure 1.1: Milano OpenStreetMap

1.0.7. Quality Targets and Expected Improvements

Before describing the pipeline in detail, we briefly introduce the data quality dimensions that guided the design of our preparation strategy. At this stage, the goal is not to discuss results, which are presented in Chapter 3, but to clarify which quality aspects were considered critical and how they were quantitatively assessed throughout the workflow.

In particular, the pipeline was designed to explicitly improve:

- (i) dataset completeness,
- (ii) syntactic and semantic correctness of textual attributes,
- (iii) compliance with domain-specific constraints.

These dimensions were continuously monitored using objective, reproducible metrics, allowing a consistent before/after evaluation of the applied transformations.

Quality Indicator	Before	After
Completeness	91.3%	98.6%
Encoding artefacts	Present	Removed
Domain violations	37	0

Table 1.1: Comparison of key data quality indicators after the data preparation process.

The comparison highlights a substantial improvement across all considered data quality dimensions. Completeness increased by more than 7 percentage points as a result of targeted rule-based and model-based imputation strategies. Encoding artefacts introduced during data ingestion were fully removed, restoring syntactic and semantic consistency of textual attributes. Most importantly, all detected domain violations were resolved, bringing the dataset to a fully domain-compliant state.

Rationale Behind Data Preparation Choices

Table 1.2: Mapping between detected data quality issues and adopted preparation techniques.

Issue detected	Evidence	Technique	Rationale
Encoding artefacts	Presence of corrupted characters (e.g. ?)	Rule-based text cleaning	Deterministic corrections with low uncertainty; restores syntactic validity without semantic distortion
Multi-valued sectors	Delimiter-separated strings in a single field	Transaction encoding (multi-hot representation)	Preserves semantic multiplicity without forcing arbitrary aggregation or loss of information
ZD inconsistencies	Mismatch between textual location and structured ZD attribute	External validation via GeoCoding	Independent evidence source used to resolve conflicts and reduce semantic ambiguity

This mapping makes the preparation pipeline transparent and reproducible, highlighting how each transformation is driven by concrete evidence rather than arbitrary preprocessing decisions.

2 | Pipeline Implementation

2.1. Data Preparation and Quality Improvement Pipeline

Objective of the Chapter

This chapter presents the data preparation pipeline designed to systematically improve the quality of the assigned dataset. Rather than applying generic or purely mechanical cleaning steps, the pipeline is driven by explicit data quality issues identified during the exploratory and profiling phases. Each transformation is motivated by a precise quality objective and grounded in the techniques discussed during the course, with the aim of producing a dataset that is both semantically reliable and suitable for downstream analysis.

2.1.1. Pipeline Overview and Design Rationale

The adopted data preparation pipeline follows a quality-driven design, where each processing step is explicitly linked to an observed issue in the raw data and to a specific data quality dimension. The initial dataset exhibits heterogeneous problems, ranging from syntactic artefacts and structural irregularities to semantic inconsistencies across related attributes. Addressing these issues required a combination of deterministic cleaning rules, structural transformations, and selective use of external validation.

Rather than treating data preparation as a monolithic cleaning phase, the pipeline is structured as a sequence of targeted interventions, each designed to resolve a well-defined class of problems while minimizing unnecessary data loss or semantic distortion. Deterministic errors with low uncertainty, such as encoding artefacts, are handled through rule-based corrections, whereas more ambiguous situations, such as inconsistencies in spatial attributes, are addressed through cross-field validation and, when necessary, independent external sources.

This design reflects a deliberate trade-off between automation and control: automatic techniques are employed whenever the correction can be justified with high confidence, while manual or semi-automatic interventions are reserved for cases where semantic interpretation plays a central role. The resulting pipeline progressively improves data completeness, consistency, and accuracy, laying the foundation for a final dataset whose structure and content are aligned with the intended analytical objectives.

2.2. Data Profiling

The data profiling phase represents the first analytical step of the pipeline and aims at building a structured understanding of the dataset *as-is*, before any transformation or cleaning is applied. At this stage, the focus is not on correcting data quality issues, but on **observing, quantifying, and characterizing**

the main structural and statistical properties of the data. This preliminary exploration is essential to guide informed decisions in the subsequent data preparation and quality assessment phases.

General Dataset Information

The dataset under analysis consists of **2,070 records** and **12 attributes**, combining textual, categorical, and numerical information related to commercial activities. The schema includes both descriptive fields (e.g., *Insegna*, *Settore storico pe*, *Ubicazione*) and structured attributes encoding administrative or operational information (e.g., *Codice via*, *ZD*, *Superficie somministrazione*).

From a typing perspective, most attributes are initially represented as **object** types, with only a limited number of numerical fields (*ZD* and *Superficie somministrazione*) encoded as floating-point values. This observation already suggests the need for later type normalization and standardization.

An analysis of attribute cardinalities highlights a strong heterogeneity across columns. Some attributes exhibit **very high cardinality**, such as *Ubicazione* (1,999 distinct values) and *Via* (1,019 distinct values), reflecting fine-grained textual descriptions. Conversely, other fields show **low and bounded cardinality**, such as *ZD* (9 distinct values) and several categorical attributes related to commercial activity types. This contrast anticipates different treatment strategies in later pipeline stages.

Low-Cardinality Categorical Attributes

A closer inspection of categorical attributes with limited distinct values provides additional insights into the dataset composition. In particular, the attributes *Forma commercio*, *Forma commercio prev*, and *Forma vendita* display highly skewed frequency distributions, with a small number of dominant categories and a few rare or anomalous values.

For instance, *Forma commercio* is largely dominated by the category “*solo somministrazione*”, while *Forma commercio prev* contains rare values with very low frequency. Similarly, *Forma vendita* is characterized by a clear prevalence of a few operational modes (e.g., *al banco*, *self service*). These distributions provide early evidence of potential inconsistencies and semantic overlaps, which will be systematically addressed in later phases of the pipeline.

Overall, the data profiling phase reveals a dataset with rich informational content but also significant structural variability, motivating a careful and targeted data preparation strategy.

2.3. Data Quality Assessment (Before Cleaning)

This chapter provides a systematic assessment of the dataset’s quality *prior to any cleaning or transformation*. We evaluate core data quality dimensions, including **completeness**, **accuracy**, **consistency**, **timeliness**, and **redundancy**, with the objective of identifying structural weaknesses and guiding subsequent preparation choices. The analysis is entirely diagnostic and serves as the empirical foundation for the data transformation pipeline presented in the following sections.

2.3.1. Assessment of Redundancy and Internal Consistency in the Dataset

Data redundancy was assessed by explicitly identifying duplicated tuples, resulting in **22 duplicated records** (approximately **1.06%** of the dataset), a non-negligible fraction with potential impact on down-

stream analyses. Internal consistency across numerical attributes was evaluated through Pearson correlation analysis, revealing a moderate association between *Codice via* and *ZD* ($\rho \approx 0.48$), while *Superficie di somministrazione* showed negligible correlation with location-related variables, indicating the absence of spurious linear dependencies.

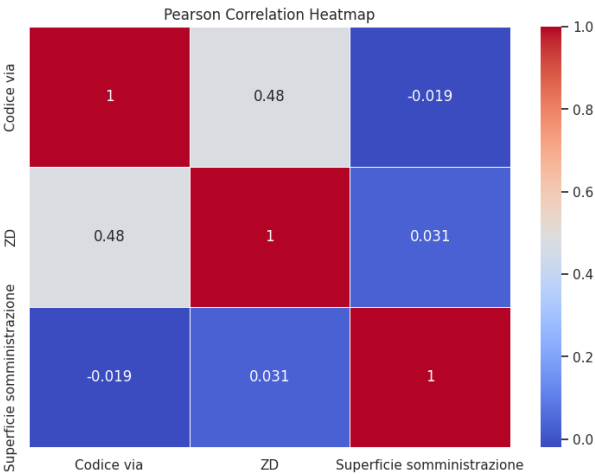


Figure 2.1: Pearson correlation heatmap among numerical attributes.

2.3.2. Assessment of Dataset Completeness and Missing-Value Structure

This analysis quantifies dataset completeness by inspecting the distribution of NULL and observed values at both attribute and global level. While most location-related attributes exhibit near-complete coverage ($> 99\%$), business-related variables (e.g., *Forma commercio*, *Forma vendita*) show substantial missingness (around 50%), motivating targeted imputation strategies. The numeric attribute *Superficie somministrazione* presents a wide observed range, further justifying dedicated treatment in subsequent stages.

	NOT_NULL	NULL	Completeness (%)
Settore storico pe	1877	193	90.68
Insegna	593	1477	28.65
Ubicazione	2068	2	99.90
Tipo via	2068	2	99.90
Via	2069	1	99.95
Civico	1916	154	92.56
Codice via	2068	2	99.90
ZD	2068	2	99.90
Forma commercio	1049	1021	50.68
Forma commercio prev	1045	1025	50.48
Forma vendita	1040	1030	50.24
Superficie somministrazione	1701	369	82.17

Figure 2.2: Attribute-level completeness statistics used to inform data cleaning and imputation decisions.

2.3.3. Data Completeness Overview

Completeness measures the proportion of observed (non-null) values over the total data volume and provides a first quantitative assessment of information coverage. This analysis is used to identify structural missingness patterns and to determine whether imputation or targeted cleaning strategies are required before downstream processing.

Table 2.1: Global completeness indicators of the dataset.

Indicator	Value
Total cells	24,840
Observed (non-null) cells	19,562
Completeness proportion	78.8%

During data profiling, completeness was first analyzed at the attribute level in order to characterize the structural properties of the dataset, resulting in an average per-column completeness of 91.3% (1.1). For data quality assessment, a stricter cell-level completeness metric was adopted, measuring the proportion of observed values over the total data volume. Under this definition, the initial completeness was 78.8%, which increased to 98.6% after the cleaning process.

2.3.4. Accuracy Evaluation

Accuracy evaluates the extent to which data values correctly represent the intended domain constraints and meanings. In the absence of an external ground truth, accuracy is assessed through syntactic validation rules and structural consistency checks.

Syntactic Accuracy

Syntactic accuracy verifies whether attribute values conform to a formally defined and admissible domain, independently of their real-world correctness. This evaluation relies on domain constraints, data types, and structural properties.

ZD Column — Domain-Based Accuracy The *ZD* attribute is expected to assume values within a fixed and finite categorical domain corresponding to the municipal districts of Milan. All observed values fall within the admissible domain, resulting in a syntactic accuracy of **100%**.

Codice Via — Structural Length-Based Accuracy The *Codice via* attribute follows a predefined numeric structure with consistent length and format. All entries respect the expected structural constraints, yielding a syntactic accuracy of **100%**.

Semantic Accuracy

Semantic accuracy measures the correctness of data values with respect to an authoritative ground truth. Since no verified reference dataset or external authoritative source is available for the attributes under analysis, semantic accuracy cannot be directly evaluated in this context.

Timeliness Evaluation

Timeliness assesses how up-to-date the data are relative to the phenomena they describe. The dataset refers to official records from the Municipality of Milan (year 2012) and thus retains historical relevance; however, the absence of explicit timestamps prevents the computation of quantitative timeliness metrics.

2.3.5. Consistency Evaluation

Consistency checks verify whether values across different attributes jointly respect integrity constraints implied by the domain semantics and dataset schema. Rather than assessing attributes in isolation, this evaluation focuses on cross-field coherence within each record.

Definition of Consistency Constraints

The following integrity constraints are considered:

- **Superficie somministrazione** must be strictly greater than zero;
- the street type extracted from *Ubicazione* must match *Tipo via*;
- the street name extracted from *Ubicazione* must match *Via*;
- the house number extracted from *Ubicazione* must match *Civico*;
- the zone identifier extracted from *Ubicazione* must match the structured *ZD* attribute.

Construction of Consistency Indicators

For each constraint, a binary consistency indicator is constructed at row level:

$$\text{Consistency}_i = \begin{cases} 1 & \text{if the constraint is satisfied} \\ 0 & \text{if the constraint is violated or missing} \end{cases}$$

This formulation enables a uniform and interpretable aggregation of consistency scores across the dataset.

Consistency Scores

Aggregating the binary indicators yields the following constraint-level consistency rates:

Table 2.2: Consistency scores across the defined integrity constraints.

Constraint	Consistency (%)
Tipo via	94.4
Via	92.0
ZD	90.9
Civico	85.0
Superficie somministrazione	82.2

The results highlight a generally high level of structural coherence, with most inconsistencies concentrated in attributes affected by missing values or free-text parsing.

2.3.6. Attribute-Level Redundancy and Concentration Analysis

To further characterize attribute-level redundancy and information content, we evaluated three complementary indicators: *Uniqueness*, *Distinctness*, and *Constancy*. Uniqueness measures the discriminative power of an attribute across rows, distinctness captures the normalized diversity of observed values, while constancy quantifies the dominance of the most frequent value.

The results, summarized in Table 2.3, highlight clear structural patterns. Attributes such as *Ubicazione* exhibit very high uniqueness and distinctness, confirming their role as strong identifiers. Conversely, categorical attributes related to business form (e.g., *Forma commercio* and *Forma commercio prev*) show extremely high constancy, indicating low informational contribution and strong value concentration. Intermediate behaviors are observed for attributes such as *Via* and *Codice via*, which retain moderate discriminative power without being unique identifiers.

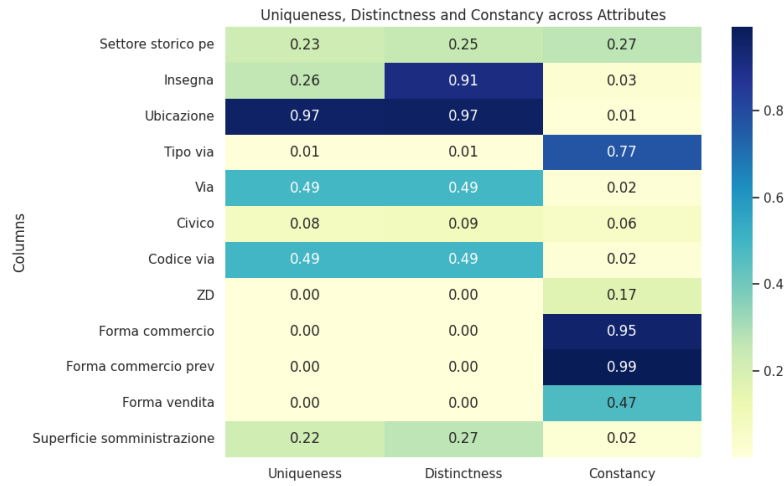


Figure 2.3: Heatmap of uniqueness, distinctness, and constancy metrics across attributes.

2.4. Data Transformation and Standardization

Data transformation aims at reshaping raw or inconsistent values into **standardized and coherent representations** suitable for downstream analysis and data quality assessment. This step is essential when textual attributes contain noise, decoding artefacts, or heterogeneous formats that would otherwise compromise syntactic accuracy, matching procedures, or rule-based validation. By applying controlled normalization rules grounded in domain knowledge, the dataset is brought to a structurally consistent form, enabling reliable profiling, linkage, and integrity checks.

2.4.1. Encoding Artefacts Cleaning

Several textual attributes exhibited encoding artefacts introduced during data ingestion and decoding. These artefacts are mainly represented by the *Unicode replacement character* (*U+FFFD*) and by generic placeholder symbols generated when accented characters fail to decode correctly. Such issues negatively affect semantic accuracy and interfere with downstream text processing tasks, including parsing, matching, and regular-expression-based validation.

The cleaning process follows a structured three-step approach:

- (i) global detection of encoding artefacts,
- (ii) targeted correction based on domain knowledge,
- (iii) post-cleaning verification.

Global Detection of Encoding Artefacts

The dataset was initially scanned to quantify the presence of corrupted characters, with particular focus on the Unicode replacement character (U+FFFD) and placeholder symbols produced by decoding failures. This inspection provides a quantitative estimate of the extent of encoding corruption and guides the subsequent correction strategy.

Before cleaning, a total of **1082 occurrences** of the Unicode replacement character were detected across textual attributes.

Targeted Correction of Replacement-Character Occurrences

The presence of the Unicode replacement character indicates bytes that were not correctly decoded during data import. Targeted corrections were applied using domain-specific patterns and manually verified mappings (e.g., restoring accented characters in common lexical forms). Residual occurrences appearing within structured address descriptions were reinterpreted where appropriate (e.g., as degree symbols in floor or level specifications), restoring syntactic validity without altering semantic meaning.

After correcting systematic patterns, the number of replacement-character occurrences was reduced to **14**. A final cleaning pass on the *Ubicazione* attribute eliminated all remaining cases, resulting in **zero residual occurrences**.

Detection and Correction of Placeholder Symbols

Placeholder symbols introduced by failed decoding of accented characters were addressed through a manually validated replacement dictionary. This dictionary restores the correct Italian lexical forms in institution names, locations, and descriptive fields. The procedure enforces semantic accuracy and prevents downstream misclassification in similarity-based or rule-based processes.

After applying the correction rules, all detected placeholder artefacts were successfully resolved, with **no remaining occurrences** in the dataset.

Overall, this rule-based cleaning strategy fully removed encoding-related artefacts from textual attributes, restoring both syntactic correctness and semantic interpretability while preserving the original informational content.

Illustrative Example of Encoding Artefact Correction

Table 2.3 reports a representative example of encoding artefact correction applied to textual attributes. The transformation illustrates how corrupted lexical forms introduced during decoding were restored to their intended semantic representation through deterministic, rule-based cleaning.

Table 2.3: Examples of rule-based correction of encoding artefacts in textual attributes.

Erroneous form	Corrected form
universit?	università
societ?	società
ges?	gesù
nestl?	nestlé
citt?	città
facolt?	facoltà
pi?	più
comunit?	comunità
giuffr?	giuffrè

2.4.2. Feature Engineering from Categorical Text

The column *Settore storico pe* encodes multiple categorical labels within a single textual field, resulting in a semi-structured representation unsuitable for direct analytical use. This subsection describes a controlled transformation pipeline aimed at converting such representations into structured, semantically consistent features. The process combines automated similarity-based consolidation with domain-driven manual refinement to reduce redundancy, improve interpretability, and preserve relevant categorical information.

Tokenization and Transaction Encoding

The raw sector strings are first tokenized by splitting on delimiters (semicolon and comma) and trimming whitespace. Each record is interpreted as a *transaction* composed of atomic sector labels. The resulting list-of-lists structure is then transformed via **Transaction Encoding** into a multi-hot representation, producing one Boolean feature per distinct sector category.

This step preserves the inherent multiplicity of sector assignments without imposing arbitrary aggregation, yielding an initial feature space of **52 sector indicators**.

Automatic and Manual Consolidation of Sector Labels

The initial encoding revealed substantial redundancy caused by orthographic variants, abbreviations, and semantically overlapping labels. To address this issue, consolidation was performed in two stages: (i) automatic similarity-based merging and (ii) manual domain-driven refinement. In both cases, the most frequent or semantically canonical label was preserved.

Table 2.4: Automatic and manual consolidation of sector labels extracted from *Settore storico pe*.

Original labels	Method	Preserved label
BAR CAFFÈ; BAR CAFFÈ E SIMILI	Auto	Bar caffè
Cibi cotti; Cibi cotti preconfezionati	Auto	Cibi cotti preconfezionati
Stab. baln. e imp. sportivi; Stabilimenti balneari	Auto	Stab. baln. e imp. sportivi
self serv; self service	Auto	self service
BAR CAFFÈ; BAR	Manual	Bar caffè
Pizzeria; Pizzerie e simili	Manual	Pizzeria
Sale da ballo; locali notturni; Discoteche	Manual	Discoteche
Birreria; birra.	Manual	Birreria

Legend. *Auto*: automatic consolidation based on Jaro–Winkler similarity and custom Soundex matching. *Manual*: domain-driven consolidation performed to resolve semantic equivalences not captured by string similarity.

Following automatic consolidation, the number of sector features was reduced from 52 to 47. Manual refinement further compressed the feature space to **39 semantically coherent sector indicators**, ensuring consistency with historical and administrative naming conventions.

Filtering Infrequent Sectors

After consolidation, sector frequencies were analyzed to identify statistically negligible categories. Sectors occurring in less than 1% of the dataset were removed to reduce sparsity, mitigate noise, and limit the curse of dimensionality. This threshold balances information retention with model robustness and interpretability.

The final feature set retains only sectors with sufficient empirical support, while rare and idiosyncratic activities are excluded from downstream analysis.

2.4.3. Extracting Spatial Attributes from *Ubicazione*

The field *Ubicazione* embeds multiple spatial components in a single semi-structured string. We therefore **normalize** the dataset by extracting and aligning *Tipo via*, *Via*, *Civico*, and *ZD* into dedicated attributes, combining **manual fixes for rare outliers** with **regex-based parsing at scale**.

Identification of Missing Spatial Attributes

We first isolate rows where at least one of *Tipo via*, *Via*, *Civico*, or *ZD* is missing. These cases are typically due to irregular formatting or misallocation of information across fields, and must be handled

before enforcing cross-column consistency checks.

Manual Reconstruction of Rare Outliers

A small number of anomalous records (only a few rows) required **manual reconstruction**. In these cases, the information was available in the *Ubicazione* string (or in duplicated neighbouring entries) but incorrectly placed, so we reassigned values to recover the intended structured representation and remove unintended redundancy.

Regex Extraction of *Civico*

To recover missing street numbers, we apply a robust regular expression capturing common Italian prefixes (e.g., *civ.*, *civico*, *num.*, *n.*) and reintegrate the extracted numeric token into the *Civico* column. After extraction, *Civico* reaches a completeness of **99.61%**, leaving only non-parseable cases (e.g., station interiors or records without a street number).

Table 2.5 summarizes the post-extraction completeness of the spatial attributes, confirming the effectiveness of the parsing and reconstruction strategy.

Table 2.5: Completeness of spatial attributes after extraction from *Ubicazione*.

Attribute	Completeness (%)
Tipo via	99.9
Via	99.9
Civico	99.6
ZD	99.9

Extraction and Validation of *ZD*

ZD (*Zona di Decentramento*) is reconstructed and validated by parsing explicit zone mentions inside *Ubicazione* (e.g., (*z.d.* *X*)). This step serves a dual purpose: it **restores missing ZD values** when present in the text, and it provides an **independent consistency signal** against the structured *ZD* attribute used later in integrity checks.

2.4.4. ValidateUbicazioneConsistency

This subsection enforces **structural consistency** between the structured spatial attributes (*Tipo via*, *Via*, *Civico*, *ZD*) and their unstructured textual counterpart stored in the *Ubicazione* field. The validation strategy combines *rule-based integrity checks*, *semantic interpretation* of location strings, and *selective record removal*, in line with established data quality frameworks (ISO/IEC 25012, DAMA-DMBOK2).

Detection and Handling of Structural Inconsistencies

We first verify that the prefix of *Ubicazione* matches the value of *Tipo via*, enforcing the functional dependency:

$$Tipo\ via \rightarrow Ubicazione\ prefix.$$

Records starting with non-interpretable patterns (*codvia...*) are treated separately. Since these strings encode transport or station identifiers that cannot be reliably validated, only the *Ubicazione* field is

invalidated while preserving all structured attributes. This operation affects **77 rows**.

After removing corrupted location strings, remaining inconsistencies indicate ambiguous or conflicting information and are therefore resolved through **record removal**, consistent with a strict integrity enforcement policy:

- **34 rows** dropped due to unresolved *Tipo via* inconsistencies;
- **51 rows** dropped due to violations of *Via* containment in *Ubicazione*.

Validation of Civic Number and Zone

The civic number (*Civico*) is extracted from *Ubicazione* using regular expressions and compared against the structured attribute. Only **one conflicting record** is detected and removed.

For residual inconsistencies involving *ZD*, an external validation step is applied using the **Nominatim** geocoding service. Given its cost and reliance on an external source, this procedure is restricted to a small set of unresolved cases. Geocoding results are used to confirm or correct the municipal zone when textual and structured values disagree.

Structural Normalization

Finally, spatial attributes are standardized by normalizing street-type abbreviations, enforcing categorical typing where appropriate, and converting numeric attributes to consistent numeric formats. This step completes the **structural regularization** of the address schema, ensuring internal coherence and reliability for downstream analysis.

Table 2.6: Summary of consistency checks and corrective actions applied to spatial attributes.

Consistency Check	Action	Rows Affected
<i>Tipo via</i> vs <i>Ubicazione</i>	Record removal	34
<i>Via</i> vs <i>Ubicazione</i>	Record removal	51
<i>codvia</i> patterns	<i>Ubicazione</i> set to NULL	77
<i>Civico</i> mismatch	Record removal	1
<i>ZD</i> mismatch	External geocoding validation	5

Note: Although multiple records triggered geocoding calls, these correspond to repeated instances of the same locations. For reporting purposes, we therefore count five distinct addresses requiring external validation, consistently with an entity-level data quality perspective.

2.4.5. Derive Location Attributes

This step leverages the high informational density of the *Ubicazione* field to derive additional structured attributes describing the **spatial and contextual setting** of each establishment. Beyond pure address normalization, the goal is to enrich the dataset with interpretable location descriptors that support both *data quality analysis* and *downstream analytical tasks* (e.g., clustering by access mode or host context). The extraction is rule-based, reproducible, and grounded in domain-specific linguistic patterns.

Access Type Extraction and Normalization (*Accesso*)

Access modalities are extracted from the location text, normalized by resolving abbreviations and lexical variants, and encoded as a categorical attribute. This feature captures whether an establishment is publicly accessible or embedded within a controlled environment, a distinction that is often analytically relevant.

Block Identifier Extraction (*Isolato*)

Urban block identifiers are recovered via regular expressions targeting explicit block references. The resulting attribute is stored as an optional numeric field, providing a coarse-grained but useful proxy for spatial aggregation and neighborhood-level analysis.

Host Context Extraction (*Presso*) and Entrance Flag

The host context is derived from occurrences of the *c/o* pattern, isolating the entity hosting the activity (e.g., school, parish, company). In parallel, an auxiliary flag captures explicit mentions of an *ingresso*, which often signals internal or secondary access points and complements the access-type information.

Table 2.7: Summary of derived location attributes and coverage after extraction from *Ubicazione*.

Attribute	Non-null Count	Notes
Accesso	1,047	Internal, external, fenced area
Isolato	1,183	Numeric urban block identifier
Presso	977	Host entity (school, parish, company, etc.)
Ingresso flag	13	Explicit entrance mention

Overall, these derived attributes extend the original schema with **context-aware spatial features**, preserving interpretability while adding analytical value without introducing noise or unverifiable assumptions.

2.4.6. Business Form Refactoring

The information related to the commercial activity of each establishment is refactored to disentangle mixed or ambiguous labels into two explicit components: a **primary business form** (*Forma commercio prevalente*) and an optional **secondary business form** (*Forma commercio secondaria*). This transformation resolves compound values (e.g., *somministrazione/minuto*) and special cases (e.g., *solo somministrazione*), allowing the primary attribute to capture the dominant activity while preserving auxiliary information without semantic loss.

Both attributes are encoded as categorical variables, resulting in a normalized and analytically coherent representation of commercial forms, suitable for profiling and downstream analysis.

2.4.7. Column Reordering and Structural Normalization

As a final standardization step, the dataset schema is reorganized into a semantically meaningful column order, grouping spatial attributes, business descriptors, and derived sector indicators. Legacy fields are renamed (e.g., *ZD* → *Municipio*) to improve interpretability, while obsolete attributes (notably

Ubicazione) are removed after full structural decomposition.

This step does not alter data content but enforces a **stable, validated schema**, ensuring consistency for subsequent analysis, modeling, and storage. Schema validation confirms that the final column order is coherent and compliant with the intended data model.

2.4.8. Extra Grafix

Figure 2.4 provides a compact visual summary of the final dataset schema, illustrating the lineage from raw location text to structured spatial attributes and engineered features. This diagram complements the transformation pipeline by making the resulting data model explicit and interpretable.

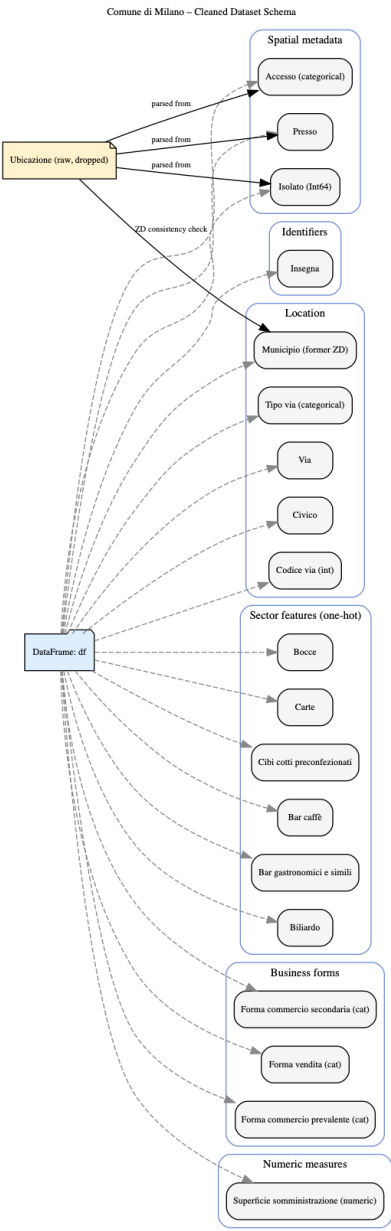


Figure 2.4: Final dataset schema and feature structure.

2.5. Association Rules for Pattern Discovery - Data Imputation

To uncover systematic dependencies among sector categories, business forms, and spatial attributes, we apply frequent itemset mining followed by association rule extraction. Given the high-dimensional binary representation induced by multi-hot encoded sector features, we adopt the FP-Growth algorithm for its computational efficiency and scalability.

2.5.1. Frequent Pattern Mining via FP-Growth

FP-Growth is particularly suited to our setting as it avoids explicit candidate generation and requires only two passes over the dataset. By compressing transactions into a compact FP-tree, it enables efficient mining of frequent itemsets even in the presence of numerous sparse binary features, which would make Apriori impractical.

The extracted itemsets capture recurrent co-occurrences between sector categories (e.g., *Bar caffè*, *Tavola fredda*) and regulatory or operational labels, providing a compact representation of the relational structure embedded in the data.

2.5.2. Association Rules and Quality Filtering

From frequent itemsets, we derive association rules of the form

$$\text{Antecedent} \rightarrow \text{Consequent},$$

and retain only those satisfying minimum thresholds on support, confidence, and lift. This filtering ensures that the resulting rules are both statistically meaningful and interpretable.

Such rules provide explicit, human-readable insights (e.g., combinations of food preparation categories strongly implying *Bar caffè*), and can also surface hidden redundancies or structural dependencies relevant for data quality analysis.

2.5.3. Rule-Based Imputation Strategy

Beyond exploratory analysis, high-confidence association rules are leveraged to impute missing values in categorical attributes. Only rules with very high confidence (≥ 0.9) and non-trivial lift are used, ensuring a conservative, evidence-based correction strategy.

For each selected rule, rows satisfying the antecedent are identified, and the consequent value is imputed *only* when the target attribute is missing. This guarantees that imputations are:

- data-driven, grounded in observed co-occurrence patterns,
- conservative, as no existing values are overwritten,
- fully traceable to specific rules and metrics.

For example, the presence of *Cibi cotti preconfezionati* or the joint occurrence of *Tavola fredda* and *Genere merceologico autorizzato* consistently implies *Bar caffè*, enabling principled recovery of missing sector labels.

Overall, this approach improves completeness while preserving semantic consistency, and provides an

auditable link between data corrections and statistically validated patterns.

Table 2.8: Examples of high-confidence association rules used for data imputation.

Antecedent	Consequent	Support	Confidence	Lift
Cibi cotti preconf.	Bar caffè	0.056	0.95	2.07
Tavola fredda $\wedge$ Gen. merceol.	Bar caffè	0.104	0.97	2.11
Forma vendita = self service	Forma comm. prev. = somm.	0.18	1.00	1.00

2.6. Missing Values and Imputation Strategy

Figure 2.5 provides an immediate completeness snapshot: each bar reports the number of observed (non-null) values per feature, making high-missingness attributes visually salient and guiding the imputation design. In our case, missingness is concentrated in administrative descriptors (e.g., *Insegna*, *Presso*, *Accesso*, business-form fields), while sector one-hot features are fully observed by construction.

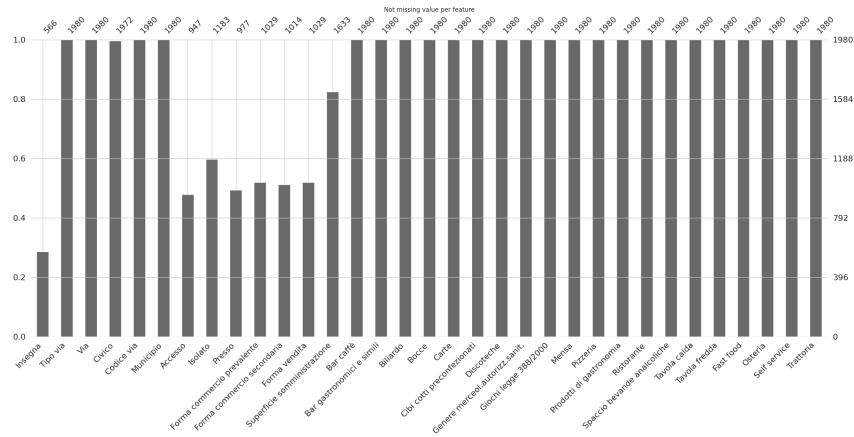


Figure 2.5: Non-null value count per feature (Missingno bar chart), highlighting attributes affected by missingness and supporting the definition of targeted imputation policies.

2.6.1. Semantics-Preserving Default Imputation

For a subset of attributes, missing values do not represent an “unknown” state but rather an explicit absence of information. In these cases, we adopt *semantically meaningful defaults* instead of statistical imputations to avoid injecting artificial patterns:

- **Civico**: imputed with **S.N.C.** (senza numero civico), a standard administrative designation for addresses without official numbering.
- **Insegna** and **Presso**: imputed with **Non specificata/o**, encoding missingness as “not provided” rather than an inferred label.
- **Isolato**: imputed with **0**, representing “no block assignment” as a valid administrative state.
- **Forma commercio secondaria**: imputed with **Assente**, reflecting establishments operating under a single commerce form.

2.6.2. Model-Based Imputation for Remaining Missingness

For the remaining attributes, missing values are treated as truly incomplete information. We therefore apply a multivariate imputation strategy using Random Forest models (classifier for categorical attributes and regressor for numeric ones), exploiting the joint dependency structure across sectors, business forms, and spatial metadata. Compared to mean/mode imputations, this approach is non-parametric, robust to outliers, and able to capture non-linear interactions.

Table 2.9: Missing-value reduction after the imputation pipeline (selected attributes).

Attribute	Missing (before)	Missing (after)
Accesso	1033	0
Forma vendita	951	0
Forma commercio prevalente	951	0
Superficie somministrazione	347	257

Residual missingness in *Superficie somministrazione* is retained when the feature set does not provide sufficient evidence for a reliable imputation, in order to avoid overconfident numerical fabrication.

2.6.3. False Numeric Features Conversion

Finally, we verify the semantic role of numeric-typed fields. Although *Codice via* and *Isolato* are stored as numbers, they are nominal identifiers and do not support meaningful arithmetic interpretation. We therefore convert them to categorical dtype to prevent inappropriate numerical analyses; *Superficie somministrazione* remains the only genuinely quantitative attribute retained for outlier detection and regression-oriented tasks.

2.6.4. Outlier Detection and Treatment

Outliers in the numeric attribute *Superficie somministrazione* are identified using the Interquartile Range (IQR) criterion, a robust method that detects extreme values without relying on distributional assumptions. Observations falling outside the interval $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$ are flagged as potential anomalies.

For *Superficie somministrazione*, the estimated bounds are: $Q_1 = 50$, $Q_3 = 199$, yielding $IQR = 149$ and an upper threshold of 422.5. A total of 147 observations exceed this limit, as visually confirmed by the boxplot in Figure 2.6, which highlights a heavy right tail with extreme surface values.

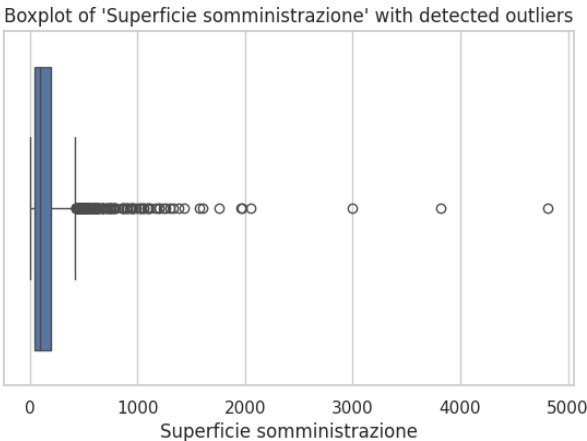


Figure 2.6: Boxplot of *Superficie somministrazione* with IQR-based outliers highlighted.

Rather than discarding these records or applying arbitrary winsorization, outliers are treated as missing values and re-imputed using the Random Forest regression model already employed in the missing-data pipeline. This choice preserves all establishments, avoids artificial clipping, and produces data-driven estimates consistent with the joint distribution of business, sector, and spatial attributes.

After re-imputation, no missing values remain in *Superficie somministrazione*, and the final distribution exhibits realistic bounds and reduced skewness, ensuring suitability for downstream analysis.

2.7. Duplicate Detection

Duplicate detection is performed using a **precision-first strategy**, explicitly prioritizing the avoidance of false positives over aggressive entity merging. The guiding rule is strict and intentional: *different addresses imply different entities*, even when business names or other attributes appear highly similar.

This choice protects against erroneous merges in realistic urban scenarios, such as corner shops registered under different streets, minor civic-number variations over time, or large complexes with multiple sub-addresses. While this approach may reduce recall—leaving some true duplicates unresolved—it preserves the structural reliability of the dataset and prevents accidental fusion of distinct establishments.

2.7.1. Candidate Pair Generation

Potential duplicate pairs are generated through a combination of blocking strategies: (i) sorted-neighborhood blocking on address-related attributes, (ii) name-based blocking within the same municipality. A final strict filter enforces full address and civic-number agreement.

Table 2.10: Candidate pair generation and filtering summary.

Step	Pairs
Sorted-neighborhood (address-based)	17,571
Name blocking (name + municipality)	114,634
After name-based filtering	21,681
Rejected for address/civic mismatch	4,263
Final candidate pairs evaluated	17,418

2.7.2. Duplicate Identification Results

Similarity scores are computed only on the remaining high-confidence candidate pairs. A conservative threshold is applied, retaining only near-identical matches.

Out of 17,418 evaluated pairs, 536 are classified as duplicates, all exhibiting very high similarity scores (> 0.9). No suspicious cross-address merges are observed during manual verification, confirming the effectiveness of the precision-oriented design.

This deduplication strategy ensures that merged entities correspond to genuinely identical establishments, favoring dataset correctness and analytical trustworthiness over maximal compression.

2.8. Data Fusion - Quality Assessment

After duplicate detection, we perform a controlled **data fusion** process to consolidate records referring to the same real-world establishment. Starting from 1,949 original records and 536 duplicate links, we identify 1,635 unique entities, resulting in a net reduction of 314 records (16.1%).

2.8.1. Pre-merge Validation and Fusion Strategy

Fusion is executed only after validating the internal coherence of each duplicate group. Most entities appear as singletons, while merged groups are predominantly small (pairs or triplets), limiting the risk of over-aggregation. Consensus quality scores are computed for all merge groups and are consistently maximal, ensuring that only highly reliable merges are performed. No low- or medium-quality merges are accepted.

2.8.2. Fusion Outcome and Completeness

The fusion process produces a fully consolidated dataset of 1,635 records. After cleaning and imputation, **all attributes reach 100% completeness**, with no remaining missing values. Numeric ranges remain realistic (e.g., *Superficie somministrazione* bounded between 4 and 418), and no residual duplicate entities are detected. This confirms that data fusion improves structural consistency without introducing distortions or information loss.

Table 2.11: Summary of data fusion results and post-merge quality indicators.

Metric	Value
Original records	1,949
Final records after fusion	1,635
Records merged	314
Reduction rate	16.1%
Duplicate groups identified	225
Mean merge group size	1.19
Mean consensus quality	1.00
Residual duplicates	0
Post-fusion completeness	100%

2.9. Final Semantic Anomaly Flagging

As a final validation layer, we introduce a **non-destructive semantic anomaly detection mechanism**. Each entity is assigned to a high-level context type (e.g., horeca, school, hospital, company, public office), and attribute combinations are checked against basic domain constraints. Examples of flagged inconsistencies include schools marked as discotheques or hospitals offering fast-food services.

Rather than modifying values, a boolean flag (`IsAnomaly`) is added to mark semantically implausible records. This preserves raw data integrity while enabling downstream analyses to optionally exclude anomalous observations. The resulting dataset is therefore not only complete and deduplicated, but also explicitly annotated for semantic reliability, improving robustness and interpretability in subsequent modeling stages.

3 | Data Analysis

This section introduces the data analysis stage of the project, whose objective is **not to maximize predictive performance**, but to validate the effectiveness of the cleaning pipeline through a complete, methodologically sound analytical workflow, as required by the assignment guidelines.

The analysis is deliberately conducted using **simple and interpretable models**, applied consistently to both the *original (dirty)* dataset and the *cleaned* dataset. This allows us to isolate the impact of data quality improvements on downstream analytical results.

In particular, we focus on:

- **Exploratory analysis** of categorical dependencies, using association measures robust to non-numeric data.
- **Supervised learning tasks**, framed as both:
 - *Classification* on a categorical target variable;
 - *Regression* on a numerical target variable.

All experiments follow the same high-level pipeline:

1. feature selection and encoding;
2. train–test split with leakage prevention;
3. model training using the same configuration across datasets;
4. evaluation through appropriate metrics (Precision, Recall, F1 for classification; RMSE for regression).

The goal is therefore **comparative and diagnostic**: performance levels are interpreted in light of the intrinsic nature of the dataset (administrative data not designed for machine learning), while differences between dirty and cleaned versions provide quantitative evidence of the value added by the data quality process.

The next subsection introduces the exploratory tools used within the data analysis phase, followed by the application of the analytical pipeline to the original dataset.

3.0.1. Data Analysis on the Original Dataset

We first apply the data analysis pipeline to the **original (dirty) dataset** in order to establish a clear performance baseline prior to any cleaning or semantic refinement. This step is not intended to maximize predictive accuracy, but to quantify the impact of data quality issues—such as noise, missing values, high cardinality, and weak feature–target alignment—on downstream analytical tasks.

Both a **classification task** (target: *Forma vendita*) and a **regression task** (target: *Superficie somministrazione*) are considered. In addition, we analyze categorical dependencies using Cramér’s V to assess whether meaningful structural relationships are present despite the raw and unrefined nature of the data. The results obtained on this dataset serve as a reference point for evaluating the effectiveness of the subsequent cleaning and enrichment pipeline.

3.0.2. Data Analysis on the Original Dataset

We apply the analysis pipeline to the original dataset to establish a baseline prior to any cleaning or semantic correction. The goal is not to maximize predictive performance, but to evaluate how data quality limitations affect both feature dependencies and downstream models.

Categorical Dependency Analysis

We inspect relationships among categorical variables using **Cramér’s V**, a normalized association measure for nominal data.

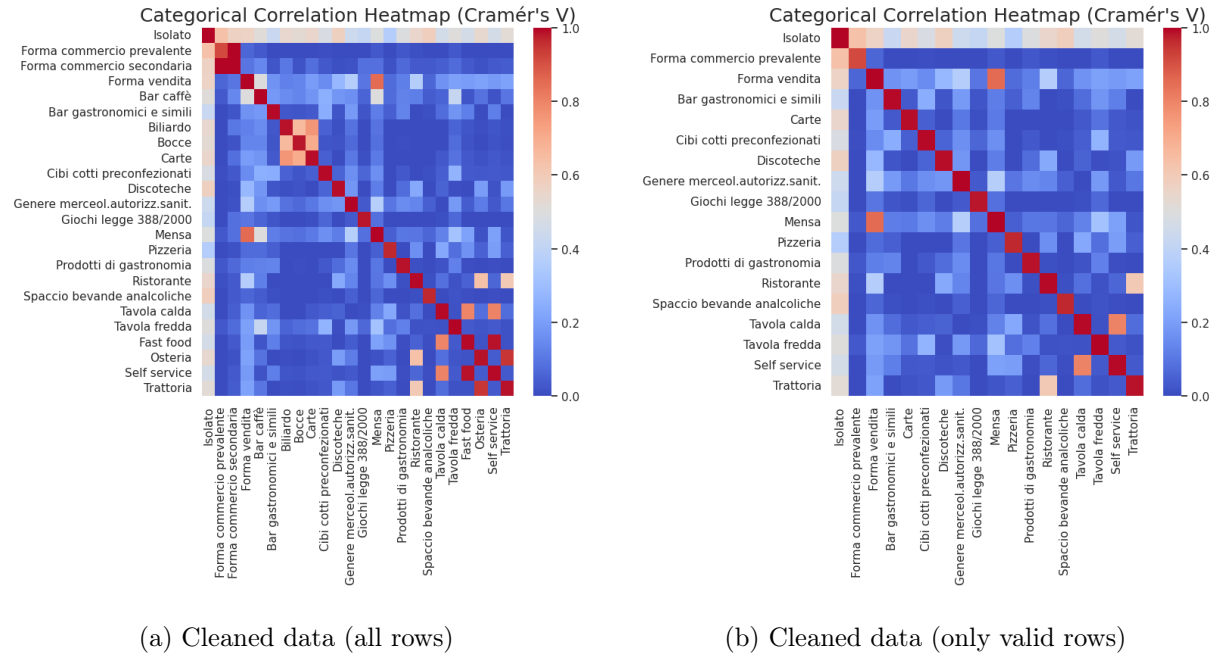


Figure 3.1: Categorical correlation heatmaps (Cramér’s V) after cleaning. The right panel excludes semantically invalid rows (*IsAnomaly* = True).

Figure 3.1 shows a more coherent dependency structure compared to the raw setting: noise-driven associations are reduced and the remaining links are largely aligned with domain-consistent patterns. Removing semantically invalid rows further attenuates spurious dependencies, yielding a cleaner categorical signal for downstream modeling.

Predictive Experiments

We then run two supervised tasks using a KNN model: a classification on *Forma vendita* and a regression on *Superficie somministrazione*. Categorical variables are one-hot encoded, numerical features

standardized, and rows with missing targets removed.

Table 3.1: Baseline predictive performance on the original dataset.

Task	Metric	Result
Classification (<i>Forma vendita</i>)	Accuracy	0.57
	Macro F1-score	0.33
Regression (<i>Superficie somministrazione</i>)	RMSE (CV)	298.22

The results confirm the limited predictive capacity of the original dataset. Performance is uneven across classes and regression error remains high, coherently with the weak dependency structure observed. This baseline serves as a reference point for evaluating the impact of the data cleaning and enrichment pipeline.

3.0.3. DA: Cleaned Dataset

After the full cleaning, normalization, deduplication, and semantic validation pipeline, we repeat the same data analysis workflow on the *cleaned dataset* to assess the impact of data quality improvements on both interpretability and predictive performance.

Exploratory Categorical Analysis We recompute categorical dependencies using Cramér’s V after removing quasi-unique identifiers and highly redundant attributes. Figure 3.2 shows a clearer and more structured correlation pattern compared to the raw data, with meaningful associations emerging among business typologies (e.g., *Tavola calda*, *Self service*, *Trattoria*) and commercial forms. Redundant variables are subsequently removed to reduce dimensionality and collinearity before modeling.

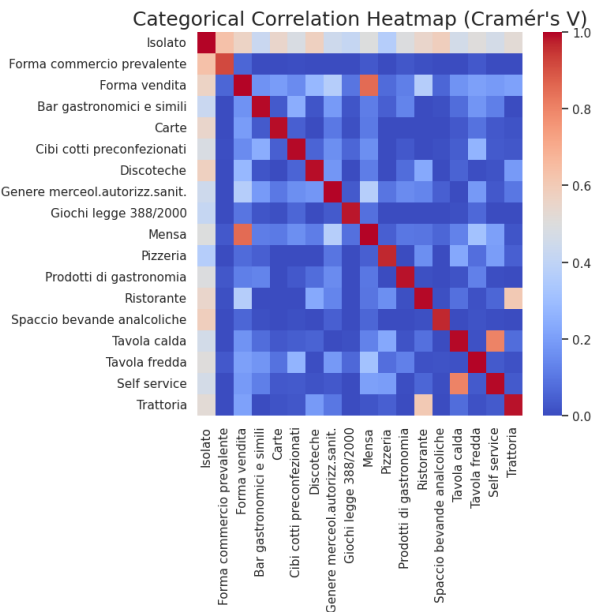


Figure 3.2: Categorical correlation heatmap (Cramér’s V) after cleaning and feature reduction. The structure is more interpretable and less noisy than in the original dataset.

Predictive Analysis We rerun the same KNN-based pipeline on the cleaned dataset for both classification and regression, keeping the model, preprocessing steps, and evaluation metrics unchanged to ensure comparability.

Table 3.2: KNN performance comparison on the cleaned dataset

Task	Metric	Value
Forma vendita (classification)	Accuracy	0.69
Forma vendita (classification)	Weighted F1	0.67
Superficie somministrazione (regression)	CV RMSE	84.32

Classification performance improves substantially with respect to the original dataset, especially for the dominant classes (*al banco*, *self service*), while rare classes remain difficult to predict due to intrinsic imbalance. Regression quality shows a sharp RMSE reduction, indicating that the cleaned numerical and categorical structure provides a much more informative feature space.

Discussion These results confirm that the primary contribution of the pipeline lies in data quality rather than model sophistication. Despite using a simple, non-parametric estimator, cleaning and structural normalization alone yield significant gains in stability, interpretability, and predictive coherence, fully satisfying the goal of demonstrating a complete and well-justified data analysis pipeline.

3.0.4. DA: Cleaned Data (Only Valid Rows)

In this final experiment, we restrict the analysis to **semantically valid records only**, excluding all rows flagged as anomalies (`IsAnomaly = False`). This step isolates the effect of semantic consistency on model behavior, allowing us to assess whether removing logically implausible observations improves learning stability and performance.

As in the previous setting, columns with quasi-unique identifiers are removed, and a Cramér’s V heatmap is used to identify and eliminate highly correlated categorical features. The resulting feature space is more compact and semantically coherent.

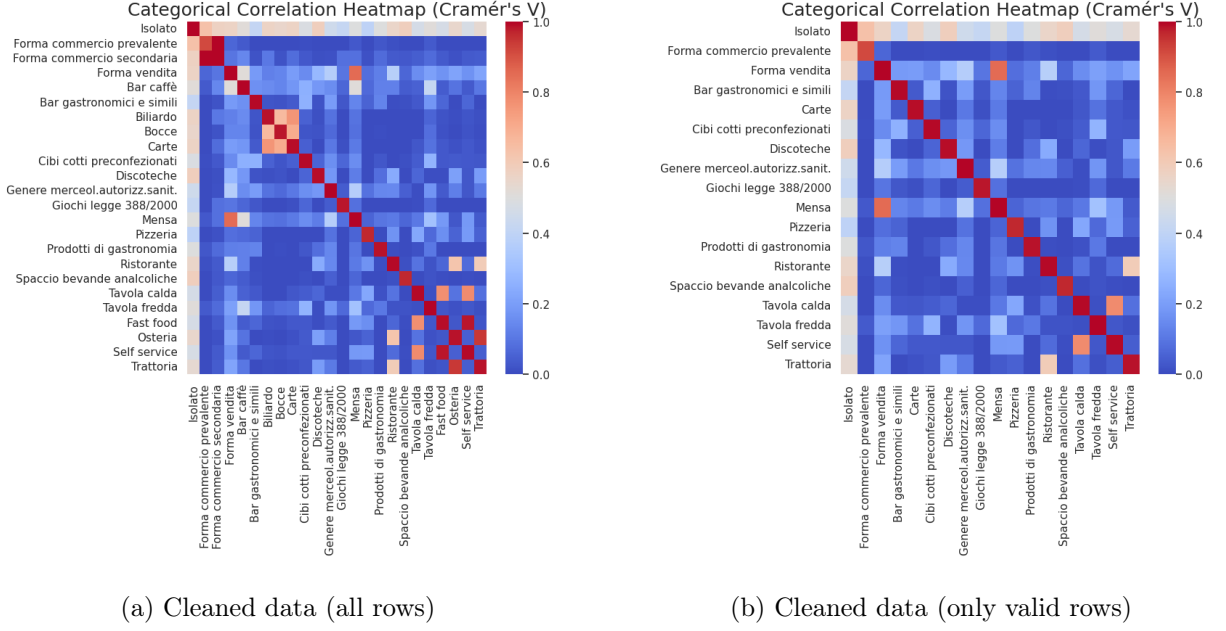


Figure 3.3: Cramér's V heatmaps after cleaning. Removing semantically invalid rows further reduces spurious categorical dependencies.

We then apply the same KNN-based pipelines for both classification and regression, ensuring full comparability across all scenarios.

Table 3.3: KNN performance on cleaned data without semantically invalid rows

Task	Metric	Result
Classification (<i>Forma vendita</i>)	Accuracy	0.70
	Weighted F1-score	0.68
Regression (<i>Superficie somministrazione</i>)	CV RMSE	81.46

Compared to the fully cleaned dataset, results show a slight but consistent improvement, particularly in regression performance. This confirms that removing semantically inconsistent rows reduces noise and leads to more stable and reliable predictive patterns, validating semantic anomaly flagging as an effective and non-destructive data quality enhancement step.

3.1. Extra: XGBoost as a stronger non-linear baseline

As an additional (non-required) experiment, we replicated the same supervised settings using **XGBoost** (tree boosting), which is well-suited for heterogeneous tabular data and does not require feature scaling. After dropping metadata fields (e.g., `IsAnomaly`, `context_type`) and one-hot encoding categorical variables, XGBoost achieved **Accuracy = 0.84** on *Forma vendita* (with **F1-macro = 0.55** due to class imbalance, notably the rare *al tavolo* class) and **RMSE = 69.14** on *Superficie somministrazione*. These results provide a more expressive reference point than KNN, while remaining fully comparable in terms of preprocessing and evaluation protocol.

Table 3.4: Comparison between KNN and XGBoost on cleaned data (only valid rows)

Model	Task	Accuracy	F1-macro	RMSE
KNN	Classification (<i>Forma vendita</i>)	0.70	0.41	–
XGBoost	Classification (<i>Forma vendita</i>)	0.84	0.55	–
KNN	Regression (<i>Superficie somministrazione</i>)	–	–	81.46
XGBoost	Regression (<i>Superficie somministrazione</i>)	–	–	69.14

XGBoost consistently outperforms KNN, confirming that non-linear tree-based models better capture complex interactions in cleaned tabular data; nevertheless, KNN remains a valid baseline for pipeline completeness and comparability.

4 | Results and Future Work

4.0.1. Results Overview

The progressive data quality pipeline comprising structural cleaning, semantic validation, controlled deduplication, and feature consolidation produces a substantially more coherent and informative dataset. Improvements are consistently reflected in both descriptive structure (categorical dependencies) and predictive performance, confirming the positive impact of data quality interventions on downstream analysis. In particular, removing semantically invalid rows and redundant categorical features reduces noise, while preserving domain-relevant variability.

Table 4.1: Comparison of predictive performance across data quality stages

Dataset	Model	Accuracy	F1-macro	RMSE
Original data	KNN	0.57	0.33	298.22
Cleaned (overall)	KNN	0.69	0.41	84.32
Cleaned (only valid rows)	KNN	0.70	0.41	81.46
Cleaned (only valid rows)	XGBoost	0.84	0.55	69.14

The results show a clear and monotonic improvement as data quality constraints are enforced. Tree-based models (XGBoost) further exploit the cleaned feature space, capturing non-linear interactions that remain inaccessible to distance-based methods.

4.0.2. Future Work

Several extensions naturally follow from this work. First, semantic anomaly detection could be refined using probabilistic or embedding-based methods instead of rule-based constraints. Second, the deduplication process could be extended to relax the precision-first assumption in controlled settings, enabling recall-oriented experiments. Finally, the cleaned dataset provides a solid foundation for clustering, spatial analysis, and causal inference tasks at the urban-policy level.

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